논문: https://doi.org/10.1145/3726302.3729937

CORONA

A Coarse-to-Fine Framework for Graph-based Recommendation with Large Language Models

박세현
LAMDA Lab in SKKU

LLM의 추론 능력을 후보 필터링에 도입하고, **Coarse-to-Fine 검색**으로 추천 시스템의 SOTA 성능을 달성했다.

Chain Of Retrieval ON grAphs (CORONA)

Large Language Models | # Graph Neural Networks | # Recommendation

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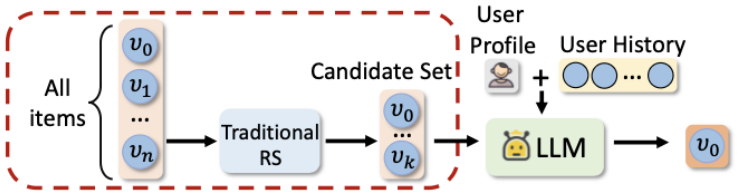
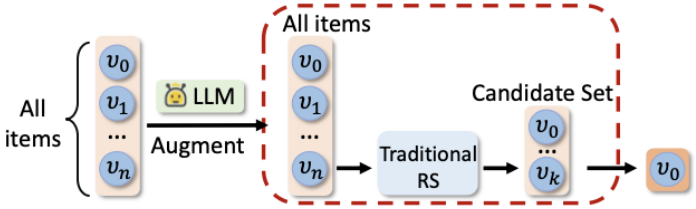
연구 배경 및 필요성

- 기존 추천 시스템의 한계
  - 후보 추출의 단순성
  - 전역 탐색의 비효율성

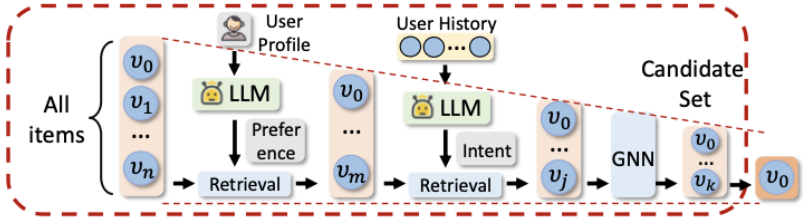
- LLM 활용의 필요성
  - 검색 단계와의 직접 결합
  - 체계적인 범위 축소



(a) Previous methods that apply LLMs after candidate filtering. These methods typically rely on traditional RSs to obtain candidate items, and use LLMs as the final predictors (*e.g.*, TALLRec [1]).



(b) Previous methods that apply LLMs before candidate filtering. These methods typically use LLMs for data augmentation, and then rely on traditional RSs to obtain candidate items (*e.g.*, LLMRec [51]).



(c) Our proposed framework using LLMs for coarse-grained retrieval and GNNs for fine-grained recommendation, directly leveraging LLMs to assist candidate filtering.

※ LLM을 '선택기' or '증강기'로써 활용이 아니라 '**추론기**'로 사용

문제 정의

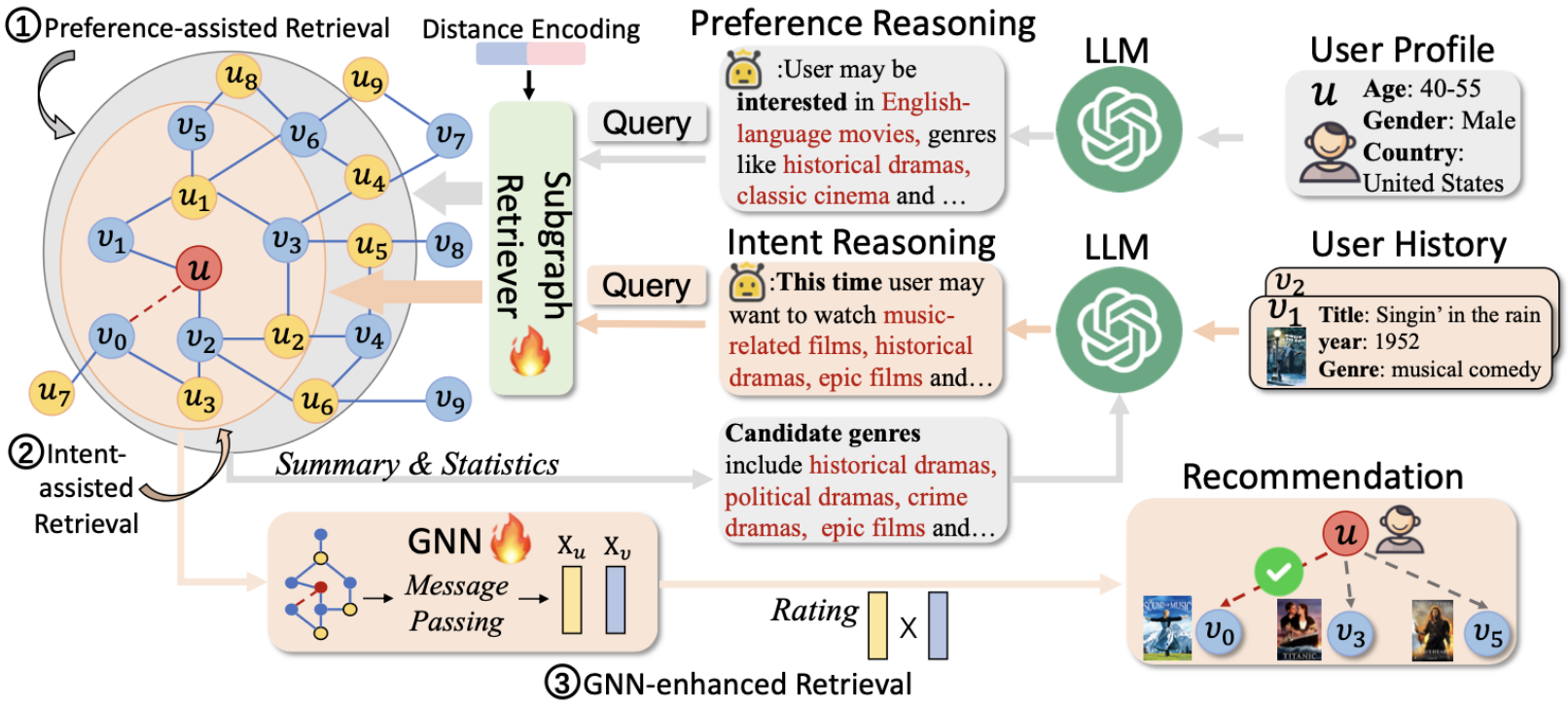
- 사용자가 다음에 상호작용할 가능성이 높은 아이템을 Top-N으로 추천
  - 사용자-아이템의 상호작용 그래프 $\mathcal{G} = (\mathcal{U}, \mathcal{V}, A)$
  - 사용자-아이템의 정보 $P_{\mathcal{U}}, T_{\mathcal{V}},\ F_{\mathcal{U}}, M_{\mathcal{V}}$
  - 사용자 이력 $L_u = \{v_1, v_2, \dots, v_n\}$

- Preference와 Intent를 반영하여 후보 서브그래프 추출 & 랭킹

연구 질문(RQ)

- RQ1. 성능이 기존 방법보다 좋은가
- RQ2. cold-start에서도 강한가
- RQ3. 각 모듈이 실제로 필요한가
- RQ4. 하이퍼파라미터에 얼마나 민감한가
- RQ5. 계산 효율성은 괜찮은가
- RQ6. LLM 추론 과정이 해석 가능성을 주는가

핵심 아이디어



- LLM은 Coarse retrieval, GNN은 Enhanced retrieval

제안 방법: CORONA

1. Preference-assisted Retrieval
   - 사용자 프로필을 바탕으로 LLM이 선호를 추론
   - 관련 서브 그래프 추출

2. Intent-assisted Retrieval
   - 1단계 결과와 최근 이력을 바탕으로 LLM이 의도를 추론
   - 더 작은 서브 그래프로 압축

3. GNN-enhanced Rerieval
   - 최종 서브 그래프 위에서 GNN 학습을 통해 아이템 랭킹

Preference-assisted Retrieval | LLM_{PR}

- Preference Reasoning
  - 사용자 프로필 정보 P_u
  - 선호도 추론 결과 텍스트 $Q_1 = \mathrm{LLM}_{PR}(P_u)$
  - 쿼리 인코딩 $E_{Q_1} = \mathrm{Encode}(Q_1)$

Module: Preference Reasoning
Instruction: Please infer the user's movie preferences based on the provided user profile. This may include preferred categories, styles, origins, or years of movies. Begin by summarizing the relevant information, and then provide your preference reasoning predictions using the user profile details along with your general knowledge.
User profile: Age: 25; Gender: Male; Country: US; Language: English; Occupation: Master student...

Preference-assisted Retrieval | LLM_{PR}

- Subgraph Retriever
  - u, u'의 거리 (hop 이웃) $\mathrm{dist}(u', u)$
  - 거리 인코딩 $e_j = \{e_1, e_2, e_3\}$
  - 특징 벡터 $\mathrm{X}_{u'} = Linear_\theta(CONCAT(F_{u'}, e_j))$
  - 서브 그래프 구성 $\mathcal{U}_1' = \mathrm{argtopk}_{\{u' \in \mathcal{U} \setminus \{u\}\}} \cos(\boldsymbol{E}_{Q_1}, \boldsymbol{X}_{u'}) \cup \{u\}$
    - 모든 아이템 포함 $\mathcal{V}_1'$

Preference-assisted Retrieval | LLM_{PR}

- Summary & Statistics
  - $\mathcal{V}_1'$의 텍스트 정보 $T_{\mathcal{V}_1'}$를 요약하고 통계 (예: 카테고리, 출시 연도 등)

$$Summary(T_{\mathcal{V}_1'})$$

Module: Summary & Statistics
Candidate Genres: The candidate items belong to the following genres: *[historical drama, political drama, epic film, music-related film, crime drama...]*
Candidate Categories: The items can be categorized into: *[independent film, documentary, animation ...]*
Years of release: The movies were released in the following periods: *[1930-1950, 1960-1970, ...]*

Intent-assisted Retrieval | LLM_{IR}

- Intent Reasoning
  - 사용자 이력
  - 의도 추론 결과 텍스트
  - 쿼리 인코딩

$$L_u$$

$$Q_2 = \mathrm{LLM}_{IR}(Summary(T_{\mathcal{V}_1'}), T_{L_u})$$

$$E_{Q_2} = \mathrm{Encode}(Q_2)$$

Module: Intent Reasoning
Instruction: Please infer the user's watching intent based on user history and candidate information. The watching intent represents the user's intention for this viewing, and you should make your selection from within the candidate range. Begin by summarizing the relevant information, then provide your intent reasoning predictions using the user history details, candidate sets, and your general knowledge.
User history: No.1: Title: The Last Emperor; Year: 1987; Genre: History drama; No.2: Singin' in the rain...
Candidate summary: Candidate Genres: ...; Candidate Categories: ...; Years of release: ...

Intent-assisted Retrieval | LLM_{IR}

- Subgraph Refinement
  - 서브 그래프 구성

$$\mathcal{U}_2' = \mathrm{argtop}\frac{k}{2}_{\{u' \in \mathcal{U}_1' \setminus \{u\}\}} \cos(\boldsymbol{E}_{\boldsymbol{Q}_2}, \boldsymbol{X}_{u'}) \cup \{u\}$$

$$\mathcal{G}' = (\mathcal{U}_2', \mathcal{V}_2', A[\mathcal{U}_2', \mathcal{V}_2'])$$

- Retriever Loss

$$\mathcal{L}1 = -\sum_{u' \in \mathcal{N}v} \left(\log \frac{\exp(E_{\mathcal{Q}1}^{\top} \cdot X_{u'})}{\sum_{u'' \in U} \exp(E_{\mathcal{Q}1}^{\top} \cdot X_{u''})} + \log \frac{\exp(E_{\mathcal{Q}2}^{\top} \cdot X_{u'})}{\sum_{u'' \in \mathcal{U}'1} \exp(E_{\mathcal{Q}2}^{\top} \cdot X_{u''})} \right)$$

  - E_{Q_1}, E_{Q_2}가 item v와 연결된 사용자들의 특징 $\mathrm{X}_{u'}$와 유사하도록 파라미터 학습 (i.e., θ and $\{e_j\}_{j=1}^{3}$)

$\mathcal{N}_v$: item v 와 연결된 users

GNN-enhanced Retrieval

- 사용자 임베딩 H_u
- 아이템 점수 $\mathrm{score}(u, v) = H_u^{\intercal} \cdot M_v$
- GNN 훈련
  - BPR (Bayesian Personalized Ranking) loss

$$\mathcal{L}_2 = -\sum_{v' \in \mathcal{V}_{\mathrm{neg}}} \log \sigma(\mathrm{score}(u, v) - \mathrm{score}(u, v'))$$

=> 최종적으로, score 기반 top-n개 추천

$\mathcal{V}_{\mathrm{neg}}$: $\mathcal{V'}_2$에서 랜덤 추출한 부정 샘플 집합

실험

- Datasets
  - Netflix / MovieLens
    - LLMRec 분할 방법
    - BERT 인코딩 $(F_{\mathcal{U}}, M_{\mathcal{V}})$
  - Amazon-book
    - RLMRec 분할 방법
    - BERT 인코딩 $(F_{\mathcal{U}}, M_{\mathcal{V}})$

- Evaluation Protocols
  - Recall (R@K)
  - NDCG (N@K)
  - K : 10, 20, 50

실험

- Methods for Comparison
  - Graph-based Collaborative Filtering Methods
    : **LightGCN, NGCF, GraphPro, GRCN**
  - LLM for Recommendation
    : **TALLRec, RLMRec, LLMRec**
  - LLM retrieval method
    : **G-retriever**

- Implementation Details
  - 128-dimensional features
  - GPT-4o-mini
  - text-embedding-ada-002

Main Result, RQ1

성능이 기존 방법보다 좋은가

Table 1: Recommendation performance on three datasets in terms of *Recall*@10/20/50, and *NDCG*@10/20/50.

Datasets	Netflix						MovieLens						Amazon-book					
Methods	R@10	N@10	R@20	N@20	R@50	N@50	R@10	N@10	R@20	N@20	R@50	N@50	R@10	N@10	R@20	N@20	R@50	N@50
NGCF	0.0357	0.0163	0.0693	0.0231	0.1090	0.0335	0.1153	0.1007	0.1305	0.1164	0.1636	0.1261	0.0844	0.0637	0.1294	0.0783	0.2151	0.1023
LightGCN	0.0385	0.0152	0.0661	0.0222	0.1252	0.0336	0.0966	0.1013	0.1314	0.1173	0.1647	0.1297	0.0857	0.0646	0.1303	0.0791	0.2154	0.1028
GraphPro	0.0397	0.0174	0.0701	0.0235	0.1263	0.0350	0.1376	0.1121	0.1462	0.1296	0.1757	0.1411	0.0865	0.0656	0.1345	0.0839	0.2203	0.1085
GRCN	0.0392	0.0169	0.0695	’0.0260	0.1318	0.0348	0.1175	0.1017	0.1356	0.1192	0.1654	0.1308	0.0862	0.0653	0.1332	0.0826	0.2179	0.1067
TALLRec	0.0498	0.0251	0.0795	0.0338	0.1303	0.0420	0.2521	0.1104	0.3522	0.1748	0.5017	0.1921	0.0857	0.0729	0.1277	0.0838	0.2420	0.1169
BinLLM	0.0502	0.0254	0.0797	0.0340	0.1312	0.0431	0.2584	0.1187	0.3657	0.1771	0.5094	0.1995	0.0972	0.0741	0.1512	0.0939	0.2451	0.1194
RLMRec	0.0504	0.0257	0.0806	0.0341	0.1314	0.0439	0.2595	0.1195	0.3668	0.1779	0.5108	0.2001	0.0968	0.0738	0.1499	0.0913	0.2432	0.1175
LLMRec	0.0526	0.0271	0.0808	0.0342	0.1317	0.0442	0.2582	0.1187	0.3528	0.1750	0.5050	0.1904	0.0862	0.0733	0.1285	0.0840	0.2424	0.1171
G-Retriever	0.0327	0.0134	0.0678	0.0213	0.1106	0.0323	0.1179	0.1052	0.1294	0.1121	0.1569	0.1225	0.0831	0.0616	0.1248	0.0719	0.2123	0.0969
CORONA	**0.0616**	**0.0279**	**0.0938**	**0.0416**	**0.1452**	**0.0487**	**0.3033**	**0.1565**	**0.4214**	**0.2017**	**0.5745**	**0.2507**	**0.1206**	**0.0855**	**0.1857**	**0.1089**	**0.3048**	**0.1299**
Improv.	17.11%	2.95%	16.09%	21.64%	10.25%	10.18%	16.88%	30.96%	14.89%	13.38%	12.47%	25.29%	24.07%	15.38%	22.82%	15.97%	24.36%	8.79%

Cold-start Result, RQ2

cold-start에서도 강한가

Table 2: Recommendation performance under item cold-start setting in terms of *Recall*@10/20/50, and *NDCG*@10/20/50.

Datasets	Netflix						MovieLens						Amazon-book					
Methods	R@10	N@10	R@20	N@20	R@50	N@50	R@10	N@10	R@20	N@20	R@50	N@50	R@10	N@10	R@20	N@20	R@50	N@50
DropoutNet	0.0175	0.0083	0.0323	0.0157	0.0414	0.0195	0.1352	0.0538	0.2107	0.1101	0.3622	0.1495	0.0425	0.0264	0.0832	0.0496	0.1482	0.1007
ALDI	0.0247	0.0099	0.0409	0.0161	0.0527	0.0202	0.1436	0.0552	0.2244	0.1102	0.3613	0.1488	0.0525	0.0307	0.0898	0.0517	0.1511	0.1026
TALLRec	0.0124	0.0075	0.0301	0.0152	0.0322	0.0181	0.1224	0.0525	0.2083	0.1099	0.3597	0.1475	0.0459	0.0278	0.0732	0.0472	0.1368	0.1004
BinLLM	0.0218	0.0086	0.0395	0.0158	0.0513	0.0200	0.01373	0.0547	0.2159	0.1095	0.3638	0.1477	0.0514	0.0296	0.0873	0.0508	0.1485	0.1012
RLMRec	0.0243	0.0096	0.0412	0.0164	0.0535	0.0207	0.1576	0.0588	0.2473	0.1105	0.3626	0.1501	0.0571	0.0319	0.0907	0.0526	0.1524	0.1035
LLMRec	0.0119	0.0073	0.0296	0.0142	0.0315	0.0175	0.1163	0.0517	0.2052	0.1095	0.3594	0.1473	0.0454	0.0273	0.0725	0.0464	0.1357	0.0971
LLM-Ins	0.0281	0.0126	0.0490	0.0182	0.0739	0.0195	0.1826	0.0854	0.2689	0.1372	0.4063	0.1665	0.0875	0.0417	0.1167	0.0704	0.1645	0.1206
CORONA	**0.0295**	**0.0130**	**0.0519**	**0.0201**	**0.0791**	**0.0217**	**0.2054**	**0.0933**	**0.2917**	**0.1425**	**0.4582**	**0.1931**	**0.0916**	**0.0453**	**0.1212**	**0.0764**	**0.1788**	**0.1314**
Improv.	4.98%	3.17%	5.91%	10.44%	7.04%	11.28%	12.49%	9.25%	8.48%	3.86%	12.77%	15.98%	4.69%	8.63%	3.86%	8.52%	8.69%	8.96%

Ablation Study, RQ3 (1)

각 모듈이 실제로 필요한가

Table 3: Combinations of different subgraph retrieval methods and GNNs on three datasets in terms of *Recall*@10/20/50, and *NDCG*@10/20/50.

Datasets	Netflix						MovieLens						Amazon-book					
Variants	R@10	N@10	R@20	N@20	R@50	N@50	R@10	N@10	R@20	N@20	R@50	N@50	R@10	N@10	R@20	N@20	R@50	N@50
GCN (full graph)	0.0307	0.0179	0.0662	0.0217	0.1012	0.0247	0.1056	0.0808	0.1371	0.0993	0.1600	0.1208	0.0786	0.0405	0.1121	0.0556	0.1845	0.0797
GCN (fixed 1-hop)	0.0319	0.0107	0.0617	0.0169	0.0981	0.0217	0.1123	0.0804	‘0.1320	0.1037	0.1599	0.1181	0.0773	0.0427	0.1133	0.0516	0.1925	0.0790
GCN (fixed 2-hop)	0.0305	0.0097	0.0632	0.0184	0.0987	0.0274	0.1092	0.0912	0.1244	0.1086	0.1495	0.1178	0.0726	0.0443	0.1135	0.0582	0.1882	0.0796
GCN (CORONA)	**0.0616**	**0.0279**	**0.0938**	**0.0416**	**0.1452**	**0.0487**	**0.3033**	**0.1565**	**0.4214**	**0.2017**	**0.5745**	**0.2507**	**0.1206**	**0.0855**	**0.1857**	**0.1089**	**0.3048**	**0.1299**
GT (full graph)	0.0302	0.0104	0.0641	0.0192	0.1008	0.0285	0.1055	0.0907	0.1235	0.1074	0.1482	0.1170	0.0719	0.0435	0.1127	0.0571	0.1854	0.0787
GT (fixed 1-hop)	0.0294	0.0081	0.0655	0.0152	0.0988	0.0342	0.1035	0.1039	0.1170	0.0828	0.1501	0.1137	0.0672	0.0330	0.1109	0.0451	0.1835	0.0787
GT (fixed 2-hop)	0.0345	0.0117	0.0665	0.0160	0.1119	0.0305	0.1044	0.0864	0.1370	0.1043	0.1463	0.1193	0.0696	0.0453	0.1187	0.0601	0.1855	0.0790
GT (CORONA)	**0.0628**	**0.0318**	**0.0931**	**0.0294**	**0.1468**	**0.0471**	**0.3072**	**0.1528**	**0.4165**	**0.1988**	**0.5504**	**0.2619**	**0.1203**	**0.1028**	**0.1928**	**0.1202**	**0.3011**	**0.1221**

Ablation Study, RQ3 (2)

각 모듈이 실제로 필요한가

Table 4: Ablation study of different components on three datasets in terms of *Recall*@10/20/50, and *NDCG*@10/20/50.

Datasets	Netflix						MovieLens						Amazon-book					
Methods	R@10	N@10	R@20	N@20	R@50	N@50	R@10	N@10	R@20	N@20	R@50	N@50	R@10	N@10	R@20	N@20	R@50	N@50
w/o Preference-assisted Retrieval	0.0576	0.0227	0.0848	0.0343	0.1295	0.0431	0.2886	0.1375	0.3873	0.1970	0.5306	0.1829	0.1112	0.0747	0.1535	0.0909	0.2228	0.1036
w/o Intent-assisted Retrieval	0.0529	0.0208	0.0785	0.0329	0.1006	0.0418	0.2694	0.1092	0.3590	0.1913	0.5129	0.1768	0.1037	0.0764	0.1643	0.0937	0.2028	0.0961
w/o GNN-enhanced Retrieval	0.0591	0.0264	0.0917	0.0359	0.1304	0.0457	0.2958	0.1434	0.4088	0.1943	0.5531	0.1905	0.1189	0.0814	0.1729	0.0968	0.2311	0.1106
w/o Preference Reasoning	0.0506	0.0187	0.0759	0.0297	0.0954	0.0386	0.2532	0.1158	0.3316	0.1741	0.4819	0.1703	0.0933	0.0653	0.1490	0.0875	0.1986	0.0887
w/o Intent Reasoning	0.0451	0.0142	0.0735	0.0264	0.0933	0.0371	0.2317	0.1173	0.3221	0.1476	0.4609	0.1659	0.0789	0.0621	0.1215	0.0839	0.1982	0.0835
Full CORONA	**0.0616**	**0.0279**	**0.0938**	**0.0416**	**0.1452**	**0.0487**	**0.3033**	**0.1565**	**0.4214**	**0.2017**	**0.5745**	**0.2507**	**0.1206**	**0.0855**	**0.1857**	**0.1089**	**0.3048**	**0.1299**

Ablation Study, RQ3 (3)

각 모듈이 실제로 필요한가

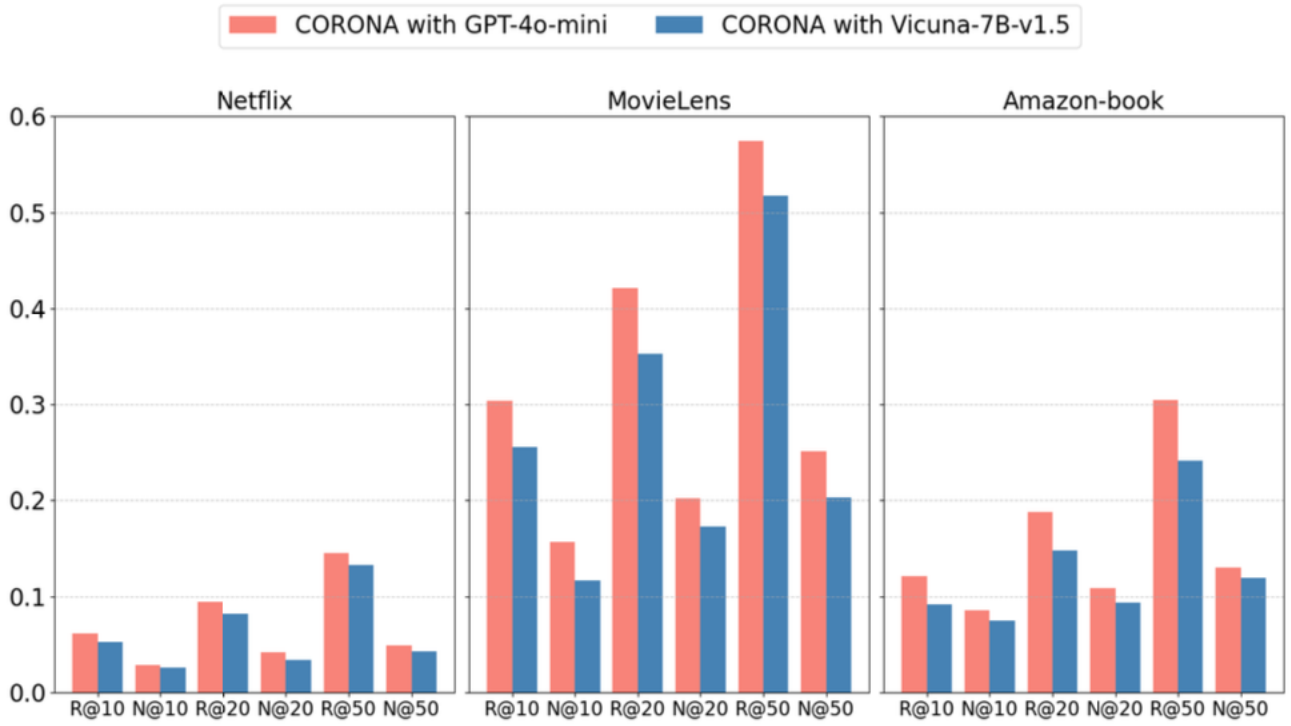


Figure 3: Applying different LLMs in CORONA on three datasets in terms of *Recall*@10/20/50 and *NDCG*@10/20/50.

Hyperparameter Analysis, RQ4

하이퍼파라미터에 얼마나 민감한가

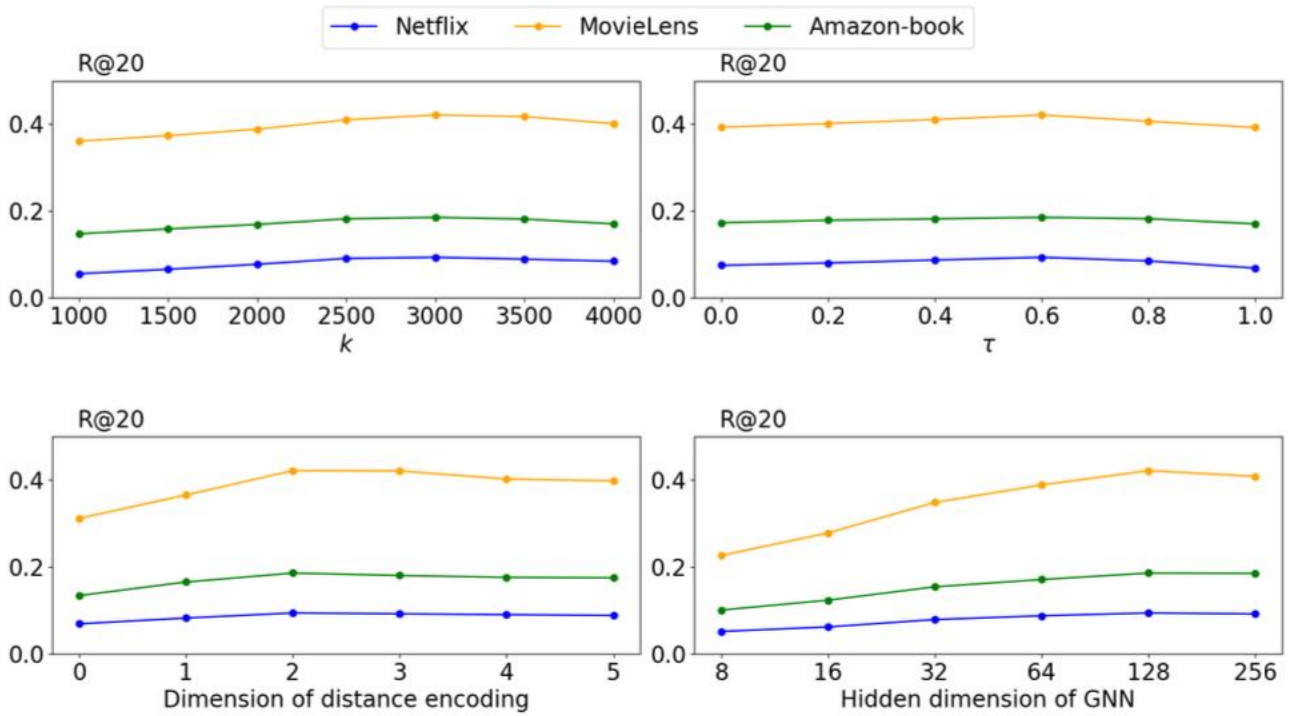


Figure 4: Hyperparameter experiments on three datasets in terms of *Recall*@20.

Efficiency Analysis, RQ5

계산 효율성은 괜찮은가

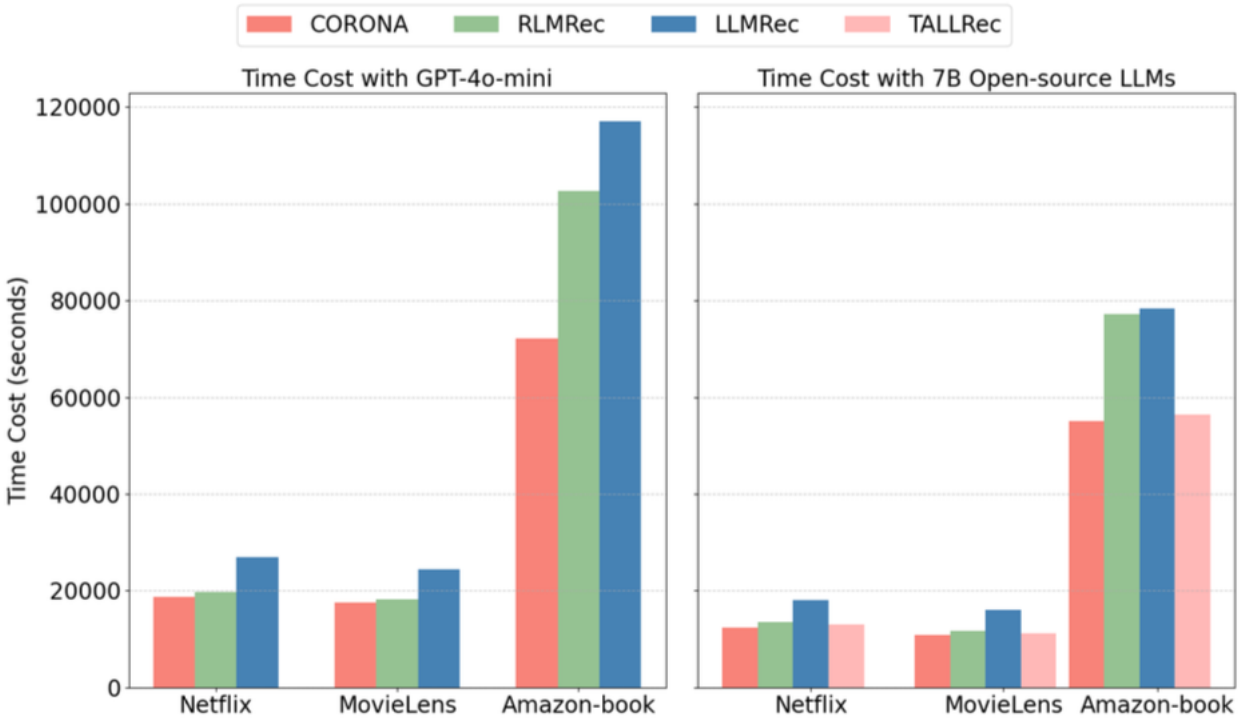


Figure 5: Total time cost (in seconds) of different LLM-based recommendation methods.

Case study, RQ6

LLM 추론 과정이 해석 가능성을 주는가

User Profile: **Age**: 40-55; **Gender**: Male; **Country**: United States; **Language**: English; **Liked directors**: Bernardo Bertolucci

Preference Reasoning: The user likely enjoys historical dramas, epic films that reflect deep social and personal themes. Prefers visually rich and thematic films (inspired by Bernardo Bertolucci). Likely favors English-language movies from the 1970s-1990s.

Summary: Candidate genres include: historical dramas, epic films, crime dramas, war films, classic cinema…

User History: **Title**: The Last Emperor ; **Year**: 1987; **Genre**: Historical Drama; **Title**: The Cook and Other Treasures; **Year**: 1970; **Genre**: Classic cinema; **Title**: Journey: 2001; **Year**: 2000; **Genre**: classic cinema; **Title**: Reds; **Year**: 1981; **Genre**: Historical Drama;

Intent Reasoning: The user's viewing preferences lean towards serious, thematic, and genre-specific films, particularly historical dramas, epic films, music-related films and international films.

Final Recommendation @5: Schindler's List (Historical drama, 1993); Crouching Tiger, Hidden Dragon (International film, 2000); Ran (International film, 1985); Lawrence of Arabia (Epic film, 1962); The Godfather (Classic cinema, 1982).

(a) Case of a male user.

User Profile: **Age**: 30-50; **Gender**: Female; **Country**: United States; **Language**: English; **Liked directors**: Ingmar Bergman

Preference Reasoning: The user likely enjoys psychological dramas and horror, artistic, moody, and emotionally complex films. The interest in European directors indicates a broader openness to international and foreign films, especially European cinema.

Summary: Candidate genres include: Horror, Psychological Thriller, Animation, Fantasy, Drama, Mystery…

User History: **Title**: Hour of wolf; **Year**: 1986; **Genre**: Horror; **Title**: Deep Red; **Year**: 1975; **Genre**: Horror; **Title**: Bambi: Platinum Edition; **Year**: 1972; **Genre**: animation; **Title**: Valerie and Her Week of Wonders; **Year**: 1970; **Genre**: Fantasy;

Intent Reasoning: The user's viewing preferences lean to something in the Horror or Thriller genres. They may also be drawn to Mystery due to its connection with psychological thrillers and horror.

Final Recommendation @5: The Shining (Horror film, 1980); Psycho (Psychological Thriller, 1960); Picnic as Hanging Rock (Mystery, 1975); The Night Porter (Psychological Thriller, 1977); The Sixth Sence (Fantasy/Horror, 1999).

(b) Case of a female user.

결론 / 기여

- Coarse-to-Fine
  - LLM을 추천 시스템에 통합하기 위한 새로운 패러다임 제시
- CORONA 프레임워크
  - 3단계 검색 프로세스로 다양한 세분화 수준에서 선택 범위를 점진적으로 정교화
- SOTA (Baseline 대비 가장 좋은 성능)
  - Recall 17.66% ↑
  - NDCG 16.06% ↑